WORKED SOLUTION

STEP III 1989, Question 1

Pure • Surfaces of Revolution

STEP III 1989, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s3-q1

The key idea

A sphere is the surface of revolution of a semicircle, so the zone's area is a standard surface-of-revolution integral. The surprise is that the integrand turns out to be constant, which is exactly why the answer depends only on the distance between the planes. In the second part the surface splits into a cone (from OP) and a spherical zone (from the arc PR). The first result then does the hard work, and the tangent condition supplies the right-angled triangle we need.

A word on the printed equation: with c^2 on the right the circle passes through O and the constant a never appears. The intended curve is clearly

$$(x-c)^2+y^2=a^2,$$

the circle with centre $C=(c,0)$ and radius a. Since $c>a$, the origin lies outside it, so tangents from O exist. We solve the question for this curve.

Solution

The area of a zone

Take a sphere of radius r with centre at the origin, and choose the x-axis perpendicular to the two planes, which are $x=h_1$ and $x=h_2$ with $-r\le h_1<h_2\le r$. The sphere is formed by rotating the upper semicircle $y=\sqrt{r^2-x^2}$ through 2π about the x-axis, so the zone has area

$$S=\int_{h_1}^{h_2}2\pi y\sqrt{1+\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2}\,\mathrm{d}x.$$

Here $\dfrac{\mathrm{d}y}{\mathrm{d}x}=-\dfrac{x}{\sqrt{r^2-x^2}}$, so for $|x|<r$

$$y\sqrt{1+\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2}=\sqrt{r^2-x^2}\sqrt{\frac{r^2-x^2+x^2}{r^2-x^2}}=\sqrt{r^2-x^2}\cdot\frac{r}{\sqrt{r^2-x^2}}=r.$$

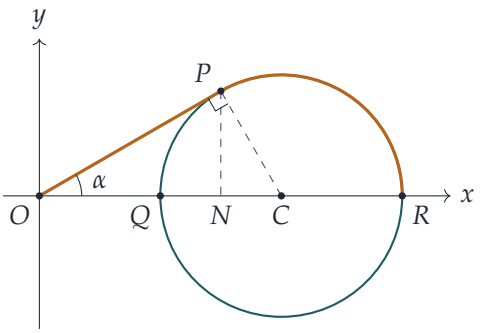
Ask yourself What happens at $x = \pm r$, where $\mathrm{d}y/\mathrm{d}x$ is infinite?

The derivative blows up at the poles, but the product $y\sqrt{1+(y')^2}$ equals r on all of $(-r, r)$, so it is bounded. If a plane is tangent to the sphere ($h_1 = -r$ or $h_2 = r$), the integral is improper only in a harmless way: integrate up to $h_2 - \varepsilon$ and let $\varepsilon \to 0$. (Alternatively, parametrise by $x = r\cos\theta$, $y = r\sin\theta$, which avoids the problem altogether.) Therefore

$$S = \int_{h_1}^{h_2} 2\pi r \,\mathrm{d}x = 2\pi r\,(h_2 - h_1),$$

which is $2\pi \times$ (radius) $\times$ (distance between the planes), as required.

The surface



Let $C = (c, 0)$ be the centre and N the foot of the perpendicular from P to the x-axis. Then $Q = (c-a, 0)$ and $R = (c+a, 0)$. The arc PR is the one that does not pass through Q (shown in amber): the other arc from P to R runs through Q and round the bottom of the circle, and rotating it would sweep out the whole sphere.

A tangent is perpendicular to the radius at the point of contact, so the angle OPC is a right angle. In triangle OPC we have $OC = c$ and $CP = a$, so by Pythagoras

$$OP = \sqrt{c^2 - a^2}.$$

Let α be the angle POC. Then $\sin\alpha = \frac{a}{c}$ and $\cos\alpha = \frac{\sqrt{c^2-a^2}}{c}$, which gives

$$PN = OP\sin\alpha = \frac{a\sqrt{c^2-a^2}}{c}, \qquad ON = OP\cos\alpha = \frac{c^2-a^2}{c}.$$

The cone. Rotating OP about the x-axis gives a cone with slant height $l = OP$ and base radius PN. Its curved surface area is $\pi \times$ (base radius) $\times$ (slant height), which is

$$\pi \cdot \frac{a\sqrt{c^2-a^2}}{c} \cdot \sqrt{c^2-a^2} = \frac{\pi a(c^2-a^2)}{c}.$$

The zone. Rotating the arc PR gives the part of the sphere (centre C, radius a) lying between the plane through N perpendicular to the axis and the tangent plane at R.

These planes are a distance

$$NR = (c+a) - \frac{c^2-a^2}{c} = \frac{ac+a^2}{c} = \frac{a(c+a)}{c}$$

apart, so by the first result the zone has area

$$2\pi a \cdot \frac{a(c+a)}{c} = \frac{2\pi a^2(c+a)}{c}.$$

Total. Adding the two pieces,

$$\begin{aligned}\text{Area} &= \frac{\pi a}{c}\left[(c^2-a^2)+2a(c+a)\right] \\ &= \frac{\pi a}{c}(c^2+2ac+a^2) \\ &= \frac{\pi a(c+a)^2}{c}.\end{aligned}$$

Checks

- Take $a=1$, $c=2$. Then $P=\left(\frac{3}{2},\frac{\sqrt{3}}{2}\right)$, the cone has area $\frac{3\pi}{2}$ and the zone 3π, so the total is $\frac{9\pi}{2}\approx 14.137$. Approximating the curve by 20 000 short segments and adding the areas of the resulting frustums gives 14.13717.
- As $a\to c$, P approaches O and the cone disappears. The surface becomes the whole sphere of radius c, and indeed $\pi c(2c)^2/c = 4\pi c^2$.
- As $c\to\infty$ with a fixed, the area grows like πac, which is the area of a long thin cone of slant height about c and base radius about a. That is what we should expect.

Common slips

- Using OC or ON instead of OP as the slant height of the cone.
- Measuring the zone's thickness from C or from Q rather than from N, the foot of the perpendicular from P.
- Using the formula $2\pi rh$ with $r=c$ rather than $r=a$: the sphere being cut is the one of radius a centred at C, not anything centred at O.

Beyond STEP

The zone result is Archimedes' theorem. Projecting a sphere horizontally onto the circumscribing cylinder preserves area, which is the idea behind the Lambert equal-area map projection. It also explains why a point chosen uniformly on a sphere has a uniformly distributed height. Areas of surfaces of revolution are the first case of the surface integral $\iint |\mathbf{r}_u \times \mathbf{r}_v|\,\mathrm{d}u\,\mathrm{d}v$ from a first-year vector calculus course. For a

beautifully geometric view of curvature and surfaces, see Needham, *Visual Differential Geometry and Forms*.

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